Public Key Cryptography, RSA Math & Web PKI Security

Mathematical foundations of asymmetric RSA encryption, Diffie-Hellman key exchange, SHA-256 digital signatures, and X.509 certificate chains.

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1. Asymmetric Cryptography and RSA

RSA security relies on the hardness of factoring large semiprimes n = p * q. Key generation uses Euler's totient phi(n) = (p-1)*(q-1) to compute public exponent e and private key d.

2. Public Key Infrastructure (PKI) & TLS

X.509 certificates bind public keys to validated domains via Certificate Authorities. Modern TLS 1.3 leverages ephemeral Diffie-Hellman (ECDHE) for Perfect Forward Secrecy.